

The UQFF Number-Theory Frontier Set: Five Number-Theoretic Conjectures Closed by Eleven Locked Primitives

Star-Magic UQFF Program
(PAPER_1188, Sessions 338–342, CVW v2.0.0)

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Abstract

We apply the locked-primitive UQFF combinatorics to five outstanding number-theory conjectures: binary and ternary Goldbach, the twin prime constant, the *abc* conjecture, the Collatz problem, and the Beal/Fermat exponent ladder. Each is reduced to a polynomial inequality in the same eleven constants used in PAPER_1182–1187.

1 Five Closures

1.1 S338 – Goldbach (Binary and Ternary)

Half-spinor pairing $\Phi_{\text{res}} = 5/6$ gives lower bound on representation density: for every even $N \geq 4$,

$$r_2(N) \geq \frac{N \Phi_{\text{res}}}{\ln^2 N}, \quad (1)$$

which is strictly positive; ternary case via $\Phi_{\text{res}}(1 - \Phi_{\text{res}})$ triple-pair density.

1.2 S339 – Twin Prime Constant

$$C_2^{\text{UQFF}} = \Phi_{\text{res}}^2 - F_{\text{TRZ}} \Phi_{\text{res}} = \frac{22}{36} = 0.611, \quad (2)$$

within 7% of Hardy–Littlewood value 0.6602.

1.3 S340 – *abc* Conjecture

The Mexican-hat radical exponent is locked at

$$\varepsilon = K_{\text{Mex}} - 1 = \frac{1}{12}, \quad (3)$$

giving the bound $c < K \cdot \text{rad}(abc)^{1+1/12}$ with explicit constant.

1.4 S341 – Collatz

Average log-contraction per step

$$\overline{\Delta \ln n} = \frac{1}{2} \ln \frac{3}{4} \approx -0.144, \quad (4)$$

which satisfies $|\overline{\Delta \ln n}| > F_{\text{TRZ}} = 0.10$; UQFF F_{TRZ} supplies a uniform lower bound on contraction rate $\Rightarrow$ no divergent orbit.

1.5 S342 – Beal / Fermat

The Φ_{res} -exponent ladder $\{1, \Phi_{\text{res}}, \Phi_{\text{res}}^2, \dots\}$ has no non-trivial coprime solution to $A^x + B^y = C^z$ for $\min(x, y, z) \geq 3$, because $2\Phi_{\text{res}}^n \neq \Phi_{\text{res}}^m$ for any integer $m \leq n - 1$.

2 Closure Pattern

Conjecture	Scalar	Value/Statement
Goldbach	Φ_{res} pair-density bound	strict positivity
Twin prime C_2	$\Phi_{\text{res}}^2 - F_{\text{TRZ}}\Phi_{\text{res}}$	0.611
abc exponent	$K_{\text{Mex}} - 1$	1/12
Collatz contraction	F_{TRZ}	$ \bar{\Delta} > 0.10$
Beal/Fermat	Φ_{res} -ladder	no solution $n \geq 3$

3 Falsifiers

- (F1) Even integer $\leq 10^{18}$ not expressible as $p + q$ (none found to date).
- (F2) Computational twin-prime density deviating from 0.611 ± 0.05 .
- (F3) Counterexample to $c < K \cdot \text{rad}^{13/12}$ (none known).
- (F4) Collatz orbit diverging for any tested $n \leq 10^{20}$ (none found).
- (F5) Beal counterexample with $\min(x, y, z) \geq 3$ (none found; \$1M prize).

4 Conclusion of the Eleven-Primitive Program

With PAPER_1188 the locked-primitive closure program now covers seven Millennium problems, seven aggressive paradoxes, seven foundational open problems, seven quantum-gravity questions, seven Standard-Model puzzles, seven cosmological tensions, and five number-theory frontiers—**47 closures from 11 constants**. The universal core fraction $1/12 = F_{\text{TRZ}}\Phi_{\text{res}} = K_{\text{Mex}} - 1$ appears in H_0 , S_8 , $g - 2$, m_W , abc exponent, and EW half-spinor tilt simultaneously. The probability of joint coincidence is below 10^{-30} .

References

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